

Yuliang Peng

ypeng12@umd.edu | [ypeng12.github.io/MyPage/](https://github.com/ypeng12) | [linkedin.com/in/ypeng12](https://www.linkedin.com/in/ypeng12) | 240-886-9716

EDUCATION

University of Maryland, College Park

M.S. in Applied Machine Learning

GPA: 3.8/4.0, Expected May 2027

B.S. in Computer Science

GPA: 3.6/4.0, May 2024

Relevant Coursework: LLM Systems, LLM Methods & Applications, Cloud Computing, Computer Vision, Deep Learning

WORK EXPERIENCE

ICBC USA — *Software Engineer Intern* | Los Angeles, CA | Jun 2026 - Aug 2026

- Integrated a vendor AI-agent platform into digital-banking workflows with Python/SQL mappings and prompt logic for deposit exceptions, card-payment issues, client-document review, and financing triage.
- Built SQL-backed pipelines to normalize 200+ client, deposit, payment, and compliance records, tracking failed states, retry attempts, audit notes, review outcomes, and escalations.
- Developed validation dashboards for recurring mobile-deposit and card-not-present payment failures, reducing manual review time by 20% and improving compliance follow-up visibility.

ModeSens — *Software Engineer* | Remote, USA | Apr 2025 - Sep 2025

- Engineered AWS/Python ingestion pipelines for 30+ retailer feeds, normalizing metadata, images, pricing, and retailer-specific schemas into reliable workflows for analytics and recommendations.
- Implemented CI/CD and validation checks in data workflows, reducing processing failures by 20% and improving reliability for analytics.
- Built image-duplicate detection with perceptual hashing, SSIM, and Selenium, eliminating 80% of reviews and improving recommendation accuracy by 25%.

Bank of China — *Software Engineer* | Los Angeles, CA | Jul 2024 - Apr 2025

- Built a transaction-monitoring backend for risk and compliance workflows, integrating SQL pipelines, exception handling, and reporting.
- Added data-quality and anomaly checks to flag inconsistent records, failed processing states, and downstream reporting gaps.
- Resolved 70+ production incidents through log analysis, root-cause isolation, and pipeline fixes, reducing blocking incidents by 20%.

All People — *Software Engineer Intern* | Remote, USA | Jan 2024 - May 2024

- Built Node.js/Express APIs and optimized SQL queries for a marketplace platform, improving transaction performance by 15%.
- Built Python ETL pipelines to aggregate user behavior data and support analytics features for recommendation workflows.

Air China — *Software Engineer Intern* | Chengdu, China | Jun 2023 - Aug 2023

- Built Spring Boot dashboards and Python data-sync scripts for flight operations, supporting crew scheduling and maintenance logs.

PROJECTS

[InferRoute](#) | LLM Inference Gateway & Control Plane | FastAPI, Redis, PostgreSQL, Docker

- Built a high-concurrency LLM inference gateway across OpenAI, Gemini, and local models with task-aware routing, Redis Pub/Sub request coalescing, timeout budgets, retries, and circuit breakers.
- Implemented prefix-hash routing with Redis-backed longest-prefix matching to improve cache locality for long-context requests and reduce redundant prefill work by 35% in repeated-prefix benchmarks.
- Developed PostgreSQL tracing dashboards and failure-injection tests for p50/p95/p99 latency, token usage, cache-hit rate, provider failover, and degraded-provider recovery.

[Quant.ai](#) | Quant Research & Backtesting Agent Platform | FastAPI, React, TypeScript, Pandas, NumPy

- Built an AI-assisted quant research platform that converts natural-language strategy ideas into backtests, signals, and risk reports using LLM tool-calling and vectorized market simulations.
- Built experiment-comparison dashboards for strategy performance, drawdown analysis, Sharpe ratio, win rate, and risk-adjusted returns.

PUBLICATION

International Cargo Charter Operations and Preighter Performance | *Transport Policy*, Vol. 184, Article 104173, 2026

- Co-authored Python route-network analysis of CAAC cargo approvals. DOI: 10.1016/j.tranpol.2026.104173.

SKILLS

ML / Data: PyTorch, TensorFlow, Hugging Face, NumPy, Pandas, Selenium

Backend / Infrastructure: FastAPI, REST APIs, Docker, Redis, PostgreSQL, AWS, CI/CD, Linux, Git

Languages: Python, C++, Java, JavaScript/TypeScript, SQL | Spoken: English, Chinese, Japanese